When Sample Selection Bias Precipitates Model Collapse

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Introduction

Recursive synthetic-data loops have two canonical baselines: the *Replace paradigm* (Shumailov et al., Nature 2024), which discards historical data and can eliminate variance, and the *Accumulate paradigm* (Kazdan et al., ICML 2025), which appends generations and is stable without biased filtering.

Theory

Proposition 1: Replace paradigm (Shumailov et al., Nature 2024)

Under the *Replace* paradigm, as $t \to \infty$,

$$\Sigma_t \xrightarrow{a.s.} \mathbf{0}, \qquad \mathbb{E}\left[\mathbb{W}_2^2(\mathcal{N}_t, \mathcal{N}_0)\right] \to \infty.$$

Replacing data eventually removes variance and drifts away.

Proposition 2: Accumulate paradigm (Kazdan et al., ICML 2025)

Under unbiased accumulation,

$$\mathbb{E}[\bar{\Sigma}_t] \to \alpha_n \bar{\Sigma}_0.$$

Thus $\mathbb{E}[\mathbb{W}_2^2] \to 2(1 - \sqrt{\alpha_n})\,\mathrm{Tr}(\bar{\Sigma}_0)$.

Unbiased accumulation keeps diversity from fully collapsing.

Theorem 1: local selection collapses diversity

With top-α selection toward a local ideal $\mathbf{u}^*$,

$$\|\bar{\boldsymbol{\mu}}_t - \mathbf{u}^*\|^2 \to 0, \qquad \bar{\Sigma}_t \to \mathbf{0}.$$

Local fidelity erases global coverage.

Local selection gains fit by erasing global coverage.

Theorem 2: diversity decays by power law

If the trajectory remains in the local basin,

$$\mathrm{Tr}(\bar{\Sigma}_t) = \mathcal{O}_{a.s.}(t^{-\psi}).$$

Tail erosion has an explicit rate.

Once trapped locally, diversity shrinks at a predictable rate.

Theorem 3: drift controls true-manifold risk

For filtered distribution $\mathcal{D}_t$ and true manifold $\mathcal{D}^*$,

$$\mathcal{R}_{\mathcal{D}^*}(h_t; g^*) \le 2\ell\epsilon\, \mathbb{W}_p(\mathcal{D}_t, \mathcal{D}^*) + \mathcal{R}_{\mathcal{D}_t}(h_t; g_t) + \mathcal{O}(\ell\delta).$$

Distribution drift upper-bounds risk on true data.

For more methodology and theoretical details, please refer to the full paper.

Synthetic data loops collapse when bias favors local fit rare modes vanish.

Scan

OpenReview paper and supplement.

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Multivariate Gaussian Modeling

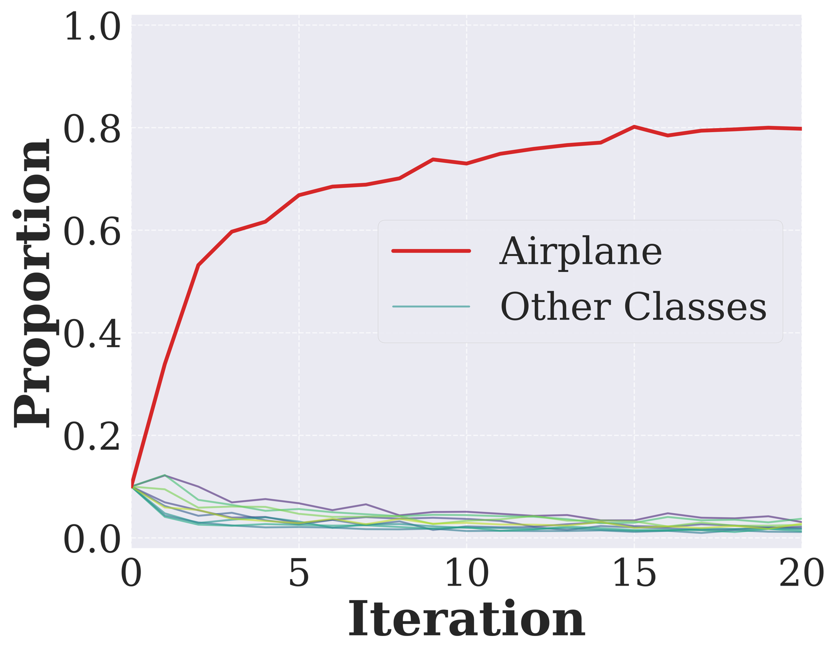
Gaussian (main text). Biased top-α selection collapses variance in both recursive baselines; accumulation slows collapse but follows the predicted power-law decay.

Anisotropic Gaussian. Direction-dependent spread still contracts when the verifier favors a restricted local region.

Gaussian mixture. Multimodal and imbalanced mixtures lose component coverage, with minority modes eroding first.

Laplace. Heavy-tailed scale contracts sharply, showing tail loss beyond Gaussian variance.

Collapse Signal



Airplane-only verification drives CIFAR-10 proportions toward Airplane, exposing systematic tail-mode pruning.

Experimental Synthesis

Across CIFAR-10, STL-10, and CelebA, skewed local references can make selection weaker than random sampling; broader distributional evidence improves FID, precision, and recall.

References

[1] Shumailov et al. AI models collapse when trained on recursively generated data. Nature, 2024.
[2] Kazdan et al. Collapse or Thrive: Perils and Promises of Synthetic Data in a Self-Generating World. ICML, 2025.

For additional experimental results, please refer to the full paper.

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